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# A Hierarchical Latent Variable Model for Ordinal Data From a Customer Satisfaction Survey With "No Answer" Responses

Eric T. BRADLOW and Alan M. ZASLAVSKY

We propose an item response theory model for ordinal customer satisfaction data where the probability of each response is a function of latent person and question parameters and of cutoffs for the ordinal response categories. This structure was incorporated into a Bayesian hierarchical model by Albert and Chib. We extend this formulation by modeling item nonresponse, coded as "no answer" (NA), as due to either lack of a strong opinion or indifference to the entire question. Because the probability of an NA is related to the latent opinion, the missing-data model is nonignorable. In our hierarchical Bayesian framework, prior means for the person and item effects are related to observed covariates. This structure supports model inferences about satisfaction of individual customers and about associations between customer characteristics and satisfaction levels or propensity to respond. We contrast this with exploratory and standard regression analyses that do not fully support these inferences. Our motivating example, an analysis of a DuPont Corporation 1992 customer satisfaction survey, is described in detail. The nonconjugate likelihood and prior prevent closed-form posterior inference. We present a Markov chain Monte Carlo solution using data augmentation. We diagnose case influence and identify outliers by importance reweighting, and apply posterior predictive model checks. The methods illustrated have application in other situations in which categorical observations can be determined by several latent variables.

**KEY WORDS:** Bayesian inference; Item nonresponse; Item response theory; Model checking; Nonignorable missing data.

## 1. INTRODUCTION

Corporations survey customer satisfaction to support crucial decisions about market segmentation, product repositioning, allocation of sales personnel, new research and development, and marketing campaigns. Given the importance and costs of these surveys, corporations should use the most effective data analysis methods.

In 1992 the DuPont Corporation conducted more than 130 surveys (one for each core business) of current and potential customers as part of its ambitious (and successful) continuous improvement program. These surveys cost millions of dollars, but the analysis methods used were restricted to calculation of means, standard deviations, and distributions of rating scores (an ordinal 1–10, "strongly disagree" to "strongly agree" scale) by item, demographic subgroup and for the overall population.

Although this information was useful as a reference, DuPont was interested in determining whether more rigorous analytical methods could provide deeper insight in the following areas.

1. Analyses of means treat the data as interval-scaled rather than ordinal. DuPont suspected that one-point differences on the extreme ends of the scales were more meaningful than those at the middle. Available analyses provided no basis for determining whether this was true.

2. The analyses did not provide an adequate basis for determining population relationships between respondent characteristics and opinions. DuPont desired an approach that could validate initial findings when true, and also determine additional associations that were not spurious.

3. The analyses did not provide stable estimates of individual satisfaction when the data were sparse due to item nonresponse.

4. The treatment of item nonresponse was of concern. Basing inferences on the mean of the observed responses makes sense only under strong assumptions that are unlikely to hold. Specifically, DuPont wanted to test whether item nonresponse was informative about satisfaction, and thus believed that it was a process of intrinsic interest.

Our goal was to address these concerns, using the following parametric models.

1. The ordinal data structure is modeled by an ordinal probit model with effects for persons and items.

2. Person and item parameters are linked to covariates through regression models, yielding hypothesis tests for covariate effects on satisfaction ratings.

3. The hierarchical Bayes framework allows for borrowing of strength across persons and items.

4. We incorporate a subject matter-based theory for item nonresponse that jointly models it with the ordinal ratings by positing a series of cognitive steps, described via latent variables, that determine the observed outcome.

Our work draws on previous research on ordinal data modeling. Item response theory models for binary data (Rasch 1960) are widely used, and were extended by Birnbaum (1968) and Lord (1980), among others. Bahadur (1961) modeled dependent multinomial responses using an

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independence model modified by a quadratic factor. Agresti (1977) and McCullagh (1980) developed generalized linear models for ordinal data. Andrich (1978), Holland (1981), and Masters (1982) discussed maximum likelihood estimation, and Albert and Chib (1993) conducted Bayesian inference using iterative simulation.

The remainder of the article is laid out as follows. In Section 2 we describe the goals, methods, and standard exploratory and descriptive analysis of a DuPont customer satisfaction survey (CSS) dataset and introduce notation. We present an analysis using standard ordinal and logistic regression in Section 3. We describe our Bayesian hierarchical approach in Section 4, and also compare inferences using the three approaches. In Sections 5.1–5.2 we identify outliers and check the influence of individual cases and of components of the model specification on the inferences. In Section 5.3 we describe diagnostics used to assess model fit and present a revised model. Finally, in Section 6 we summarize our conclusions and suggest directions for future research. In the Appendix we detail one of the three computational approaches used.

## 2. GOALS, METHODS, AND STANDARD ANALYSIS

### 2.1 Survey Description

We present an analysis of a 1992 CSS from a DuPont Corporation strategic business unit (SBU) in the polymers sector with sales of more than 1 billion dollars. DuPont had four broad goals in conducting their CSSs. The degree to which an analysis method addresses these goals largely determines its value to the company, together with ease of application and explanation.

DuPont was interested in satisfaction levels overall and for subgroups, seen as critical for short-term strategies involving customer follow-up, reallocation of resources, and other considerations. A second area concerned “no answer” (NA) responses, which management believed could be informative about underlying opinions, indicating respondents or subgroups who should be informed about product features. Third, rankings identify the most satisfied and the least satisfied clients and the most poorly rated items, which helps direct quality improvement and public relations efforts. Fourth, DuPont needed to know how confidently the direction and strength of various relationships could be established.

The SBU92 CSS, like many business-to-business surveys, used a disproportionate stratified sample oversampling larger clients and was conducted by a third-party vendor without identifying the sponsor. The survey was initially conducted by mail, with telephone follow-up. A direct contact name was used to improve response rates and collect respondent-level information. From 120 current and prospective customers, 102 persons responded (85% response), representing corporations that accounted for more than 90% of this SBU's sales. We treat all respondents as independent, ignoring the fact that a few corporations had multiple respondents. (The correlation of responses from the same corporation would be difficult to estimate from these few cases and would be unlikely to have much affect

on the conclusions.) To protect DuPont proprietary information, the name of the SBU and specific product and business lines are not specified.

The questionnaire comprised 20 items of the form: “How satisfied are you with your supplier's performance for — on a 1–10 scale, where 1 is extremely dissatisfied and 10 is extremely satisfied?”. The questions were grouped into six categories: product quality (questions 1–4), technical support (5–7), marketing support (8–9), supply and delivery (10–16), innovation (17–19), and price (20). A “no answer” (NA) category was not offered, but was coded when a person did not respond.

In addition, four covariates were fully observed for each respondent: job type, DuPont share of total polymer material purchases, type of product purchased, and type of business. DuPont considered these the most important customer attributes and included them in all 130 CSSs. The covariate levels are given in the first two columns of Table 1.

We constructed three additional covariates. One consisted of equally spaced ascending values, representing a trend in ratings by question administration order. DuPont was concerned that responses to later questions would be lower due to respondent burden. Two other covariates assessed the relationship between job type and possession of the knowledge necessary to answer items. Executive, marketing, and purchasing job types were categorized as “sales oriented,” as were the marketing and supply/delivery question domains. Manufacturing and technical job types were regarded as “technically oriented,” as were the quality, technical support, and innovation questions. One domain expertise covariate indicated that a technical person was answering a technical question; another, that a sales person was answering a sales question.

This comprised the SBU92 dataset, for which we introduce the following notation. Let  $\mathbf{Y}$  be the  $J \times K$ -dimensional satisfaction rating matrix ( $J = 102$ ,  $K = 20$ ),

Table 1. Distributions of SBU92 by Respondent Categories

|               | $n$ | $\bar{y}$ | % NA | $\beta_\alpha$ | $\beta_\zeta$ |
|---------------|-----|-----------|------|----------------|---------------|
| Share         |     |           |      |                |               |
| Major         | 41  | 8.40      | 2.0  | .431           | .881          |
| High          | 22  | 8.58      | 6.0  | .590           | .337          |
| Low           | 18  | 7.96      | 5.6  | .069           | .493          |
| Noshare       | 21  | 8.15      | 13.8 | Ref            | Ref           |
| Product       |     |           |      |                |               |
| A             | 13  | 8.22      | 2.3  | -.264          | .420          |
| B             | 5   | 9.02      | 1.0  | .489           | 1.333         |
| C             | 78  | 8.30      | 7.1  | Ref            | Ref           |
| D             | 6   | 7.61      | 1.7  | -.630          | .502          |
| Job type      |     |           |      |                |               |
| Executive     | 23  | 8.24      | 4.1  | .076           | 1.447         |
| Manufacturing | 10  | 8.34      | 1.0  | .169           | 2.083         |
| Marketing     | 4   | 8.29      | 10.0 | 0              | .960          |
| Purchasing    | 44  | 8.54      | 5.8  | .404           | 1.261         |
| Technical     | 21  | 7.91      | 9.5  | Ref            | Ref           |
| Business      |     |           |      |                |               |
| Segment I     | 17  | 8.34      | 2.1  | -.215          | .244          |
| Segment II    | 85  | 8.31      | 6.7  | Ref            | Ref           |

NOTE: Number of cases ( $n$ ), observed means ( $\bar{y}$ ), NA response rate (% NA), and model-based estimated posterior medians ( $\beta_\alpha$  and  $\beta_\zeta$ ) of coefficients for person opinion and saliency scores. Ref denotes the reference category for the dummy variable coding; that is, a coefficient fixed at 0.

where  $y_{jk} \in \{1, \dots, 10, \text{NA}\}$ . Let  $\mathbf{X}$  denote the fully observed  $J \times P$ ,  $P = 4$  covariate matrix,  $\mathbf{X}_\gamma$  the item order covariate, and  $\mathbf{X}_s$  the  $JK \times$  two-dimensional domain expertise covariate matrix, where  $\mathbf{x}_{s(jk)1} = 1$  for technical expertise,  $\mathbf{x}_{s(jk)2} = 1$  for sales expertise, and 0 otherwise.

## 2.2 DuPont's Standard Analysis

The "standard analysis" performed in 1992 by a third-party vendor for DuPont's Corporate Marketing and Business Research Division cost roughly \$30,000 and consisted of 75 pages of cross-tabulations. The main conclusions are summarized here and identified for later reference.

The marginal distribution of the responses indicates the overall level of satisfaction with DuPont's SBU92. Most of the responses (79.0%) are either 8, 9, or 10 ("satisfied"). A smaller proportion (12.6%) are from 4 to 7 ("somewhat satisfied"), and 2.5% are less than 4 ("unsatisfied"). These satisfaction levels were lower than those seen in many other DuPont businesses.

NA responses were 5.9% of all responses. Only 66 of the 102 persons answered all 20 items. One respondent (case 96) answered only two of 20 items, with mean 3.5 (the lowest observed). Eleven respondents had mean ratings below 7. DuPont concluded that (a) the percentage of NA responses is low, and (b) there is a nontrivial proportion of unsatisfied customers.

Several tabulations are summarized in Table 1, which shows the number of respondents ( $n$ ), average of the non-NA ratings ( $\bar{y}$ ), and percentage of NA responses (%NA) for subgroups. Conclusions included (c) Noshare (0% purchases) respondents have a very high NA rate (13.8%), (d) High share (70%–99%) customers give higher mean ratings than Major (100%), followed by Noshare and finally by Low (1%–69%), (e) customers who buy Product D give low mean ratings (7.61) and Product B customers have high mean ratings (9.02), with both groups having very low NA rates (1.7% and 1.0%), and (f) Technical respondents give the lowest mean ratings and have the highest NA rate (9.5%).

Inference (c) suggests that saliency affects propensity to give an NA response. This suggests that (d) means that companies take Major share customers for granted, concentrating on satisfying High share customers, whereas Low share customers may have negative opinions that restrict their purchases. Inference (e) suggests that customers with strong opinions (positive or negative) are more likely to respond. Inference (f) suggests that technically-oriented customers tend to be more critical, and recognize the limits of their knowledge.

Mean scores by item indicated areas of relatively high and low satisfaction. Innovation items have the lowest mean responses, suggesting that (g) DuPont was not perceived as innovative. Also, (h) high variability in item means indicated that customer satisfaction varied by item domain. All items had less than 15% NAs except question 14; questions 9 and 20 had only 1 out of 102 NAs. Generally, (i) there is substantial variability in NA rates by item.

The limitations of this analysis are enumerated in Section 1. Some, but not all, of these concerns are addressed in the next section.

## 3. ORDINAL AND LOGISTIC REGRESSION ANALYSIS

As a step toward a fully model-based analysis, we fitted a logistic regression model to the NAs and an ordinal logistic regression model to the ratings, conditional on a non-NA response. Each response was predicted by some combination of item covariates, person covariates, and interactions.

Three specifications were available for the person covariates: the null model of no person effects, the set of person covariates  $\mathbf{X}$ , and the saturated model with person indicator variables. Similarly, for items we could include no item effects, item order covariate  $\mathbf{X}_\gamma$ , or item dummies. This gave a total of nine combinations. Finally, we could either enter or omit domain expertise interactions  $\mathbf{X}_s$ ; these were tested only when the corresponding marginal effects were included.

As item responses by the same person are correlated, we fit the NA models with logistic regression software for correlated responses, the `svylogit` procedure in Stata (Stata Corporation 1997). Stata's `ologit` procedure for ordinal regression does not accommodate correlated data, so we estimated design effects to correct `ologit` hypothesis tests using dichotomous logistic regressions with responses  $> 8$  scored as 1. Because the results are generally consistent with those from the hierarchical models in Section 4.4, we present the results in broad terms only.

The model for ordinal responses indicated that the item order effect was insignificant, but the set of item dummies was highly significant ( $p < .0001$ ), indicating item-to-item variability [inference (h)]. The regression coefficients for  $\mathbf{X}$  were also highly significant ( $p \approx .0005$ ), even after correcting for design effects averaging around 4, suggesting significant variation in satisfaction scores across subgroups [inferences (c), (d), and (e)]. The saturated specification for persons and items significantly improved model fit ( $p < .0001$ ), but the model became numerically unstable. The two domain interactions ( $\mathbf{X}_s$ ) did not have significant effects ( $p = .402, .133$ ).

In modeling NAs, the item order variable had no significant effect, but the model with saturated item effects had significantly better fit than the model without these effects ( $p \approx .018$ ). The person covariates also had significant predictive power, corroborating inferences (c) and (f). Models with the full set of person effects were fit to 36 of 102 persons who gave at least one NA; no NAs would yield infinite parameter estimates. This saturated model did not significantly improve on the fit. The two domain expertise interactions both had highly significant coefficients ( $p < .0001$ ), although these were nearly collinear with the main effects.

These results elucidated factors affecting response, but substantial limitations remained relative to DuPont's goals. Perhaps most important, the model can estimate the effects of person covariates or effects for individuals, but not both. Consequently, the same model cannot be used for

estimating general relationships and the rankings of individuals. Moreover, although approximately valid standard errors and hypothesis tests are available using robust variance estimators for clustered data, this approach is inadequate when there is more than one random component. For example, we might treat both items and persons as random and test for significant relationships between item covariates and customer satisfaction. Furthermore, the robust variance estimation approach is not helpful for modeling relationships between individual satisfaction levels and propensity to respond. The limitations of these approaches are addressed by the hierarchical models presented in the next section.

#### 4. BAYESIAN HIERARCHICAL ANALYSIS

We demonstrate how our model, designed specifically to address DuPont's goals, can support all of these inferences while also representing the data structure more faithfully than the other analyses. We define three levels of probability models. First, the probabilities of individual responses of persons to items are modeled as a function of person- and item-specific parameters  $\theta_1$  and general parameters  $\theta_2$ ; that is,  $[Y|\theta_1, \theta_2]$ , where  $[A|B]$  denotes the density of random variable  $A$  given  $B$ . At the second level, the distribution  $[\theta_1|\theta_2]$  is modeled. At the third level, prior distributions  $[\theta_2]$  are specified. The covariates described in Section 2.1 appear at stages I and II. Inferences were derived using samples from the parameter posterior distributions to compute point and interval estimates. Samples were obtained

from three different Markov chain Monte Carlo samplers (see the Appendix).

##### 4.1 Level I: Response Model $[Y|\theta_1, \theta_2]$

The responses are modeled as conditionally independent,

$$[Y|\theta_1, \theta_2, \mathbf{X}] = [Y|P(\theta_1, \theta_2, \mathbf{X})] \\ = \prod_{j=1}^J \prod_{k=1}^K p_{jk}(y_{jk}; \theta_1, \theta_2, \mathbf{X}). \quad (1)$$

The content of the model lies in the specification of  $p_{jk}(y_{jk}; \theta_1, \theta_2, \mathbf{X})$ . We specify  $y_{jk}$  as a deterministic function of latent continuous variables (LCVs) whose distributions depend on  $\theta_1$  and on one component of  $\theta_2$ . The LCVs, and hence  $y_{jk}$ , are independent (conditional on  $\theta_1$  and  $\theta_2$ ) and distinct for each person-item pair. We define LCVs  $s_{jk}$  as latent saliency,  $t_{jk}$  as latent opinion, and  $d_{jk}$  as latent responsiveness propensity for person  $j$  and item  $k$ .

Response  $y_{jk}$  is determined by the following process, illustrated in Figure 1. Person  $j$  is confronted with item  $k$ . If the item is not salient ( $s_{jk} < 0$ ), she then gives an NA response ( $y_{jk} = \text{NA}$ ), indicating lack of knowledge or interest. Otherwise,  $s_{jk} \geq 0$ , and if she feels strongly positive ( $t_{jk} \geq C_U$ ) or strongly negative ( $t_{jk} < C_L$ ), where  $C_L$  and  $C_U$  are cutoffs, then she gives an ordinal response; a strong opinion will always be voiced. If

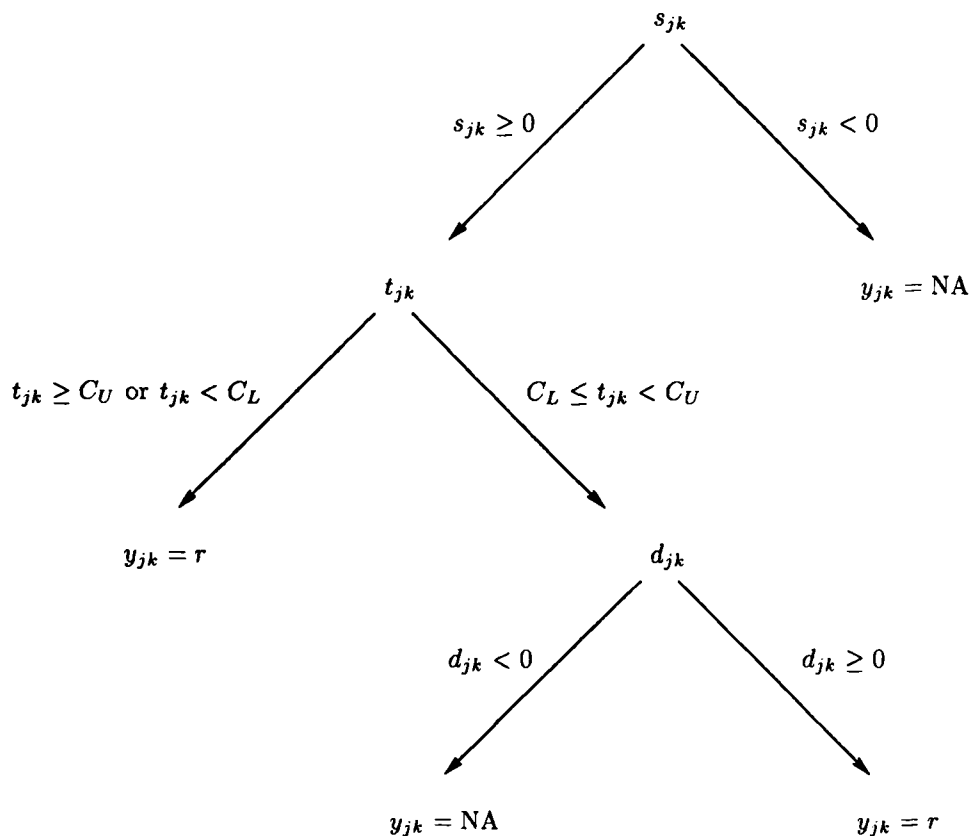


Figure 1. Latent Tree Diagram for a Single Response. The observed data are  $y_{jk}$ 's, and the latent continuous variables are  $t_{jk}$ ,  $d_{jk}$ , and  $s_{jk}$ ; cutoffs  $c$ ,  $C_L$ , and  $C_U$  are components of general parameter  $\theta_2$ .

the item is salient and  $t_{jk}$  is in an indifference zone (IZ),  $C_L \leq t_{jk} < C_U$ , she then gives an ordinal response if responsive ( $d_{jk} \geq 0$ ) (i.e. willing to express even a mild opinion), and otherwise gives an NA. The ordinal responses are determined by the latent opinion variable  $t_{jk}$  and ordered cutoffs  $c_r$ ,  $-\infty = c_0 < c_1 < \dots < c_R = \infty$  as  $r = r(t_{jk}, c)$  for which  $c_{r-1} \leq t_{jk} < c_r$ .

We set the IZ cutoffs ( $C_L, C_U$ ) at cutpoints ( $c_{r_L-1}, c_{r_U}$ ) corresponding to predetermined response levels. In the SBU92 CSS,  $r_L = 4$  and  $r_U = 7$  ("somewhat satisfied" customers), based on common practice at DuPont. The rationale for this model is that indifferent respondents have motives to not rate items, such as unwillingness to exert the effort required to generate a meaningful rating, or fear of offending the supplier or diluting opinions about other questions.

The LCVs  $s_{jk}$ ,  $t_{jk}$ , and  $d_{jk}$  are specified as independent unit normal variables,  $s_{jk} \sim N(\mu_{s(jk)}, 1)$ ,  $t_{jk} \sim N(\mu_{t(jk)}, 1)$ , and  $d_{jk} \sim N(\mu_{d(jk)}, 1)$ . The means are functions of  $\theta_1, \theta_2$ , and covariates  $\mathbf{X}$ . In the linear model  $\mu_{s(jk)} = \zeta_j + \mathbf{x}'_{s(jk)}\beta_s$ ,  $\zeta_j$  is a person-specific saliency intercept (allowing individuals to differ in knowledge and interest),  $\mathbf{x}'_{s(jk)}$  is a vector of saliency covariates for person  $j$  on question  $k$ , and  $\beta_s$  is a coefficient vector. In the SBU92 CSS,  $\mathbf{x}_s$  consists of the two-item domain by job-type interactions.

In the linear model  $\mu_{d(jk)} = \mathbf{x}'_{d(jk)}\beta_d$ ,  $\mathbf{x}_{d(jk)}$ 's are responsiveness covariate including an intercept, and  $\beta_d$  is the corresponding coefficient vector. We use all covariates to predict responsiveness ( $\mathbf{X}_d = \mathbf{X}$ ).

In the additive model  $\mu_{t(jk)} = \alpha_j + \gamma_k$ ,  $\alpha_j$  is a person effect (general satisfaction of a person across items), and  $\gamma_k$  is an item effect (general level of satisfaction on an item across all people). These are analogous to examinee ability and item difficulty in the Rasch model, widely used in psychometrics. The  $\alpha$ 's and  $\gamma$ 's are of primary interest, as we rank individuals and items based on the distributions of these parameters.

Because  $\Pr(s_{jk} < 0) = \Phi(-\mu_{s(jk)})$ ,  $\Pr(d_{jk} < 0) = \Phi(-\mu_{d(jk)})$ , and  $\Pr(c_{r-1} \leq t_{jk} < c_r) = \Phi(c_r - \mu_{t(jk)}) - \Phi(c_{r-1} - \mu_{t(jk)})$ , with  $\Phi$  the normal cumulative density function, the model can be expressed in closed form as a combination of probit models. Combining the LCV specifications with the latent tree (Fig. 1), we obtain

$$p_{jk}(r) = \begin{cases} \Pr(s_{jk} < 0) \\ \quad + \Pr(s_{jk} \geq 0, c_{r_L} \leq t_{jk} < c_{r_U}, d_{jk} < 0) & \text{if } r = \text{NA} \\ \Pr(s_{jk} \geq 0, c_{r-1} \leq t_{jk} < c_r) & \text{if } r > r_U \text{ or } r < r_L \\ \Pr(s_{jk} \geq 0, d_{jk} \geq 0, c_{r-1} \leq t_{jk} < c_r) & \text{if } r_L \leq r \leq r_U \end{cases}$$

$$= \begin{cases} \Phi(-\mu_{s(jk)}) + (1 - \Phi(-\mu_{s(jk)})) \\ \quad \cdot (\Phi(c_{r_U} - \mu_{t(jk)}) - \Phi(c_{r_L} - \mu_{t(jk)})) \\ \quad \cdot \Phi(-\mu_{d(jk)}) & \text{if } r = \text{NA} \\ (1 - \Phi(-\mu_{s(jk)})) \\ \quad \cdot (1 - (\Phi(c_r - \mu_{t(jk)}) - \Phi(c_{r-1} - \mu_{t(jk)}))) & \text{if } r > r_U \text{ or } r < r_L \\ (1 - \Phi(-\mu_{s(jk)})) \\ \quad \cdot (\Phi(c_r - \mu_{t(jk)}) - \Phi(c_{r-1} - \mu_{t(jk)})) \\ \quad \cdot (1 - \Phi(-\mu_{d(jk)})) & \text{if } r_L \leq r \leq r_U. \end{cases} \quad (2)$$

#### 4.2 Level II: Distributions of Person and Item Parameters, $[\theta_1, \theta_2]$

Vector  $\theta_1 = (\alpha, \gamma, \zeta)$  contains random effects whose prior distributions are governed by  $\theta_2$ :

$$\alpha_j \sim N(\mathbf{x}'_{\alpha(j)}\beta_\alpha, \sigma_\alpha^2), \quad (3)$$

$$\gamma_k \sim N(\mathbf{x}'_{\gamma(k)}\beta_\gamma, \sigma_\gamma^2), \quad (4)$$

and

$$\zeta_j \sim N(\mathbf{x}'_{\zeta(j)}\beta_\zeta, \sigma_\zeta^2), \quad (5)$$

where  $\mathbf{x}_{\alpha(j)}$ ,  $\mathbf{x}_{\gamma(k)}$ , and  $\mathbf{x}_{\zeta(j)}$  are covariates for person  $j$  or item  $k$ . This induces a regression structure for the predictions of  $\theta_1$  based on covariates  $\mathbf{X}$  and hyperprior parameters. The combination of (5) and the model for  $\mu_{s(jk)}$  allows saliency to depend on both person covariates and person-by-item interactions. We use all person covariates in each part of the model ( $\mathbf{X}_\alpha = \mathbf{X}_\gamma = \mathbf{X}$ ). The  $\alpha_j$  and  $\gamma_k$  correspond roughly to the coefficients of person and item dummies in Section 3, and  $\beta_\alpha$  and  $\beta_\gamma$  correspond roughly to the coefficients of person and item covariates.

#### 4.3 Hyperpriors for Parameters, $[\theta_2]$

Vector  $\theta_2 = (\beta_s, \beta_d, \beta_\alpha, \beta_\gamma, \beta_\zeta, \sigma_\alpha^2, \sigma_\gamma^2, \sigma_\zeta^2, c)$  comprises general parameters that govern the level I and II distributions. Hyperpriors for  $\theta_2$  were chosen to contain little information so that posterior distributions are proper but determined primarily by the data. Our hyperpriors are conjugate to level II distributions, multivariate normal for slopes ( $\beta_\alpha \sim N(0, a_\alpha I)$ ,  $\beta_\gamma \sim N(0, a_\gamma I)$ , and  $\beta_\zeta \sim N(0, a_\zeta I)$ ) and inverse chi-square for variance components ( $\sigma_\alpha^{-2} \sim \chi_{g_\alpha}^2$ ,  $\sigma_\gamma^{-2} \sim \chi_{g_\gamma}^2$ , and  $\sigma_\zeta^{-2} \sim \chi_{g_\zeta}^2$ ). The prior distribution of  $c$  is uniform with order constraints;  $c_7$  is fixed at 0 for identifiability. We chose hyperprior variances  $a_\alpha = a_\gamma = a_\zeta = 4$ , which restricts most of the mass of the covariate effects to  $\pm 6$  on the probit scale. The prior distributions of the variance components had  $g_\alpha = g_\gamma = g_\zeta = .5$ .

#### 4.4 Bayesian Inferences

We present initial results in roughly the same order as the standard analyses in Section 2.2, followed by additional inferences and significance tests that do not have obvious counterparts in the Section 4 and 5 analyses. For example, posterior draws can be used (as in Albert and Chib 1995 and Johnson 1997) to calculate posterior probabilities of interesting events and to indicate which inferences are most certain (posterior probabilities near 1). We also present inferences derived from the LCVs that have no counterparts in the earlier analyses. Dependencies in the data, such as correlated responses within subject, are directly represented in our model and hence do not necessitate ad hoc adjustments for clustering effects as in Section 3. Furthermore, both person and covariate effects are accommodated in the same model structure; consequently, probability inferences can be drawn for rankings of individual respondents (Laird and Louis 1989).

**4.4.1 Subgroup Inferences.** The last two columns of Table 1 shows posterior medians for the person opinion and saliency coefficients  $\beta_\alpha$  and  $\beta_\zeta$ . We use this information to compare standard and model-based analyses for the subgroup inferences (c)–(f). The analyses agree when higher NA rates are associated with lower values of  $\beta_\zeta$  and higher values of  $\bar{y}$  are associated with higher values of  $\beta_\alpha$ . Inference (a), indicating subgroup variability in opinions and saliency, is supported by the high variability in  $\beta_\alpha$  and  $\beta_\zeta$ . Inference (c), the lower propensity to respond of Noshare customers, is supported by the finding that  $\beta_\zeta$  for Major, High, and Low share customers are greater than 0 (Noshare, the reference category) with posterior probabilities  $p = .977, .715$ , and  $.881$ . The magnitude of these probabilities suggests that these effects are not spurious. In the analysis for saliency, the association between higher NA rates and higher  $\beta_\zeta$  holds for all covariates, except for the reversal of the Marketing and Technical categories.

For opinions, inference (d) regarding ratings by share is partially supported. We are fairly confident that the coefficients for High and Major share are greater than the coefficient for Low share ( $p = .983, .942$ ). The evidence that the coefficient for High is greater than that for Major share is weaker ( $p = .733$ ). Whereas the Low share had lower mean ratings than Noshare, the posterior probability that the coefficient for Low is higher than that for Noshare is  $.618$ , indicating that the minimal information for the Low share versus Noshare contrast pointed in the opposite direction from the comparison of sample means. The evidence for the ordering of opinion ratings for Executive and Marketing job types was also inconclusive ( $p = .510$ ) but opposite to that based on sample means.

Most of the model-based subgroup comparisons agreed in direction with those based on sample means, although the evidence for some of them was too weak to draw any conclusions. In three of the latter cases, the evidence from the model was opposite to that from sample means. This occurred because of the distinction between two nonresponse mechanisms; in an analysis without the IZ part of the model

(Sec. 5.3), these reversals do not occur. When a group of cases has many NAs and the data suggest that opinions may have been in the IZ, the NAs are interpreted by the model as due likely to unresponsiveness rather than to low saliency. This raises estimated saliency and lowers (because opinions in the IZ are below average) estimated opinions for the group.

Inference (e) regarding Product B customers is supported by the large probability that Product B has higher ratings (greater  $\beta_\alpha$ ) than all other products ( $p = .878$ ) and higher saliency (greater  $\beta_\zeta$ ,  $p = .615$ ). This indicates stronger evidence that Product B customers are more *satisfied* than other subgroups than they are more *knowledgeable*. Inference (f) regarding the low opinion and saliency for Technical persons is supported by substantial evidence for saliency ( $p = .645$  that it has the lowest  $\beta_\zeta$ ) but weaker evidence for opinions ( $p = .241$  that it has the lowest  $\beta_\alpha$ ). This suggests that DuPont might target Technical customers for knowledge improvement but not for satisfaction.

**4.4.2 Person and Item Inferences.** Figure 2(a) contains a plot of the 102 person means  $\bar{y}_j$  against posterior median opinion effects  $\bar{\alpha}_j$ . Plotted case numbers indicate persons whose ranking under the two measures differs by more than five positions, 26 of the 102 cases. For example, respondent 96 gave the lowest mean rating (3.5) but gave only two ordinal ratings, and has the sixth-lowest  $\bar{\alpha}$ . Because there is little information about this subject's satisfaction level, relatively little weight is given to her responses and more goes to the prior mean that characterizes the entire population. This analysis gives lower follow-up priority than the other analyses to respondent 96.

Figure 2(b) plots the 20 item means  $\bar{y}_k$  and posterior median item effects  $\bar{\gamma}_k$ . Rankings by means and posterior medians are very similar; we would expect little shrinkage, because every item has at least 83 non-NA responses. Figure 2(b) supports inference (g) and the logistic regression results regarding DuPont's perception as noninnovative; items 17–19 have the lowest  $\bar{\gamma}_k$ . Inference (h) is supported by the large variability in  $\bar{\gamma}$  across items, ranging from a low of  $-1.52$  (item 17) to a high of  $.51$  (item 16). Furthermore, the 95% posterior interval  $(.135, .368)$  for  $\sigma_\gamma^2$  indicates a high probability of substantial between-item variability.

Figure 2(c) compares two approaches to determining which individuals have the most negative opinions. Means  $\bar{y}_j$  are plotted against the posterior probability  $p_j$  that  $\alpha_j$  is ranked in the bottom 10%; case numbers are plotted where  $p_j > .50$ . The 10 cases with the highest  $p_j$  have the lowest observed means, but the ordering is different. Notably, case 96 has the lowest mean, but because it has only two non-NA responses, whether it is in the bottom 10% is uncertain ( $p_j = .75$ ). Other inferences about ranks are easily implemented using draws of the person (or item) parameters.

**4.4.3 Additional Model-Based Inferences.** Responsiveness when indifferent is latent and thus not directly observable in the standard or ordinal regression results. We found that  $\Pr(\beta_{d, \text{Major}} < \beta_{d, \text{Low}}) = .950$  and  $\Pr(\beta_{d, \text{High}} < \beta_{d, \text{Low}}) = .983$ , indicating that Low share persons are more likely than High and Major share persons to respond

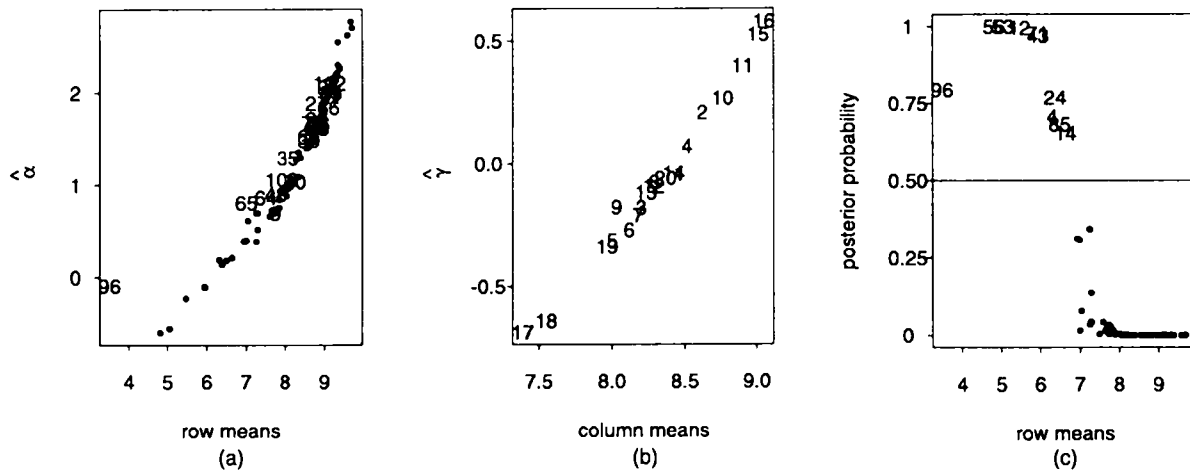


Figure 2. Sample and Posterior Estimates for Person and Item Effects. (a) Plot of the 102-person sample means  $\bar{y}_j$  versus estimated posterior median person opinion effects  $\bar{\alpha}_{j\cdot}$ ; (b) corresponding plot for the 20 item means  $\bar{y}_{\cdot k}$  and posterior median item opinion effects  $\bar{\gamma}_k$ ; (c) plot of the 102-person sample means  $\bar{y}_j$  versus posterior probability of each case being in the bottom 10% in terms of overall rating. The horizontal line indicates 50% posterior probability. Each point is plotted using its case (item) number when highlighted and  $\bullet$  otherwise.

when indifferent. This is consistent with the notion that smaller customers want to voice their opinions. Moreover, because all components of  $\beta_{d,Share}$  appear to be positive ( $p = .937, .648, .917$  for Major, High, Low), we further concluded that Noshare persons (the reference category) tend to be less likely to respond when indifferent ( $p = .535$  that it is the lowest). This suggests increasing the sample size for prospective customers (Noshare) in future studies to achieve a desired number of responses. By job type, Manufacturing persons tend to be the least likely to respond when indifferent ( $p = .740$ ).

Another set of model-based inferences concerns predictors of saliency. There is strong evidence that the saliency coefficients are positive— $\Pr(\beta_{s,1} > 0) = .970$  and  $\Pr(\beta_{s,2} > 0) = .992$ —indicating that respondents find questions more salient when they fall within their primary expertise. This confirms the logistic regression results on the domain expertise interactions and is evident in the data; the NA rate is higher when a technical person answers a sales question (9.3%) or vice versa (6.3%) than when a person answers a question within his or her expertise (5.0% in both cases). Furthermore,  $\Pr(\beta_{s,1} < \beta_{s,2}) = .925$ , indicating a stronger effect for the technical area than for sales. This might occur because specific knowledge is needed to answer technical questions, whereas general familiarity might suffice to answer sales-related items.

## 5. MODEL SENSITIVITY

### 5.1 Case Influence Analysis and Outlier Detection

The influence of each observation  $y_j$  on the posterior means of parameters is the difference between the mean  $\bar{\theta}$  based on  $[\theta|Y]$  and the case-deleted mean  $\bar{\theta}_{-j}$  based on  $[\theta|Y_{-j}]$ , where  $Y_{-j}$  is  $Y$  excluding the row for respondent  $j$ . Cases for which some components of  $\bar{\theta} - \bar{\theta}_{-j}$  are large are considered *influential*. Tracing the influence of respondents with distinctive data patterns helps us to understand and explain the workings of a complex model. Alternative

influence analyses of items or of individual item responses are not pursued here.

Using draws from  $[\theta|Y]$ , we can generate  $\bar{\theta}_{-j}$  by importance reweighting with weights

$$\frac{[\theta|Y_{-j}]}{[\theta|Y]} = \frac{[Y_{-j}|\theta][\theta]}{[Y|\theta][\theta]} = [y_j|\theta]^{-1}.$$

Computing these weights requires no extra programming, because  $[y_j|\theta]$  is simply the contribution to the likelihood from respondent  $j$ . A detailed case influence analysis for the SBU92 data and a discussion of alternative importance weights have been given by Bradlow and Zaslavsky (1997). Therefore, we only summarize the results briefly.

Five influential cases were identified. Several cases with unusually low or high mean ordinal scores were influential on the intercept for opinion scores,  $\beta_{\alpha 1}$ . Cases with an unusually high number of NAs were influential on the saliency intercept,  $\beta_{\zeta 1}$ . One case had many scores in the IZ and only one NA; deleting this case decreased the intercept for responsiveness  $\beta_{d1}$  and, paradoxically, increased the saliency model intercept to maintain the predicted number of NAs. The effects of these influential cases were consistent with our understanding of the model.

A related but distinct analysis (Zaslavsky and Bradlow 1997) concerns detection of outliers. The data for a respondent, for an item, or for a single item response could be an outlier, defined as an observation that is extreme (in the value of some statistic) relative to its predictive distribution under the model. In particular, respondents were detected that were outlying in one or more of the following senses: (1) a large number of NAs from a respondent (three respondents), (2) an unusually low mean rating (three respondents), (3) a narrow spread across the items (five respondents, two of whom answered all items identically), and (4) a wide spread across items (two respondents who marked both 1 and 10). Because outliers represent response patterns that are not well described by the model, they suggest areas in which modifications of the model should be considered, even though most of the 13 outliers (including one

that appeared twice) were not very influential on parameter estimates. We might model the data as a mixture with a large component for our main model and smaller components for some of the idiosyncratic response patterns that were observed, such as use of only the extremes or only a single response level. Because these patterns are unusual, estimation of their prevalence and of parameters for the distribution of responses in the corresponding components of the mixture should be undertaken in context of a combined analysis of several similar surveys, rather than the single survey analyzed here.

## 5.2 Sensitivity to Opinion Score Model Specification

We consider the sensitivity of parameter estimates to the Gaussian specification for latent opinions  $t_{jk}$ . This distribution was chosen for computational convenience (conjugacy of  $t_{jk}$  and  $\theta_1$ ). The logistic link is more commonly used, having a natural interpretation corresponding to the proportional odds model (McCullagh 1980). We investigated the logistic link, rescaled by a factor of  $4/\sqrt{2\pi}$  to equate the slopes of the normal and logistic cdf's at 0. This yields parameter estimates for  $\alpha_j$  and  $\gamma_k$  that are on comparable scales for either link.

Parameter estimates using the logistic link can be obtained by importance weighting with weights proportional to the ratio of (2) evaluated using the logistic and normal cdf's. The largest of the normalized weights was .0095, implying that estimates were not overly influenced by a small number of the draws.

The ranks of person, item, and saliency effects do not change much under the logistic model; rank correlations are .986, .956, and .915 for  $\alpha$ ,  $\gamma$ , and  $\zeta$ . This suggests that choice of link has little effect on the inferences of interest, which is reassuring.

## 5.3 Posterior Predictive Model Checks and Model Expansion

Posterior predictive model checks (Rubin 1984) may be used to examine the adequacy of the model. Let  $S(\mathbf{Y})$  be any univariate statistic. A posterior check assesses model fit by referring  $S(\mathbf{Y})$  to the distribution of  $S$  in replicate datasets  $\mathbf{Y}^*$  sampled from the posterior predictive distribution  $[\mathbf{Y}^*|\mathbf{Y}]$ . Model deficiencies are revealed when  $S(\mathbf{Y})$  is extreme with respect to  $[\mathbf{Y}^*|\mathbf{Y}]$ . The model can then be improved to address the deficiency.

For our model, we can conduct a full posterior predictive check, one of several mixed predictive checks (Gelman, Meng, and Stern 1997, p. 754), or a prior predictive check (Box 1980). The check that we found to be most informative uses posterior draws of  $\theta_2$  but new draws of  $\gamma$ ,  $\alpha$ , and  $\zeta$ , and corresponds to drawing both a new sample of subjects and a new set of items. This mixed check tests the adequacy of both the response model (level I) and the specification of the random respondent and item effects (level II).

Out of six posterior predictive check test statistics that we tested, two produced interesting results. An IZ check addresses whether a simpler NA model without the IZ (i.e., modeling saliency but not responsiveness) would fit as well

as the full model. Let  $S_5(\mathbf{Y}) = p_1(\mathbf{Y}) - p_2(\mathbf{Y})$ , where  $p_1(\mathbf{Y})$  is the proportion of NA responses from respondents with at least one score in the IZ and  $p_2(\mathbf{Y})$  is the same proportion for other respondents. The observed  $S_5(\mathbf{Y}) = .016$  lies at the 53rd percentile of the reference distribution, consistent with the model. To test the fit of a simpler model, we refit the model without the IZ. Then  $S_5(\mathbf{Y})$  lies at the 82nd percentile of the modified distribution, suggesting that the IZ has some explanatory power for NAs but the simpler model could not be conclusively rejected.

Check  $S_6(\mathbf{Y})$ , the maximum interitem correlation based on available (non-NA) cases over all pairs of items, detects surprisingly strong relationships between items. The correlation between items 8 and 9 gave  $S_6(\mathbf{Y}) = .862$ , which was larger than all 200 draws (maximum = .657) from the predictive check distribution, indicating lack of fit. This result was not unexpected, as the items were designed to fall into six domains (Sec. 2.1).

To model within-domain correlation, the model for  $t_{jk}$  was modified to include an effect  $\kappa_{j,Q(k)}$  for person  $j$  on items contained in domain  $Q(k)$ ,  $\mu_{t(jk)} = \alpha_j + \gamma_k + \kappa_{j,Q(k)}$ , with  $\kappa_{j,Q(k)} \sim N(0, \sigma_\kappa^2)$ ,  $\sigma_\kappa^2 \sim \chi_{g_\kappa}^{-2}$ , and  $g_\kappa = .5$ , as before. The augmented model includes the 274 original parameters and 613 additional parameters (612  $\kappa_{j,Q(k)}$  and one variance component). The fit of check statistics  $S_1$ – $S_5$  is essentially unchanged. The observed  $S_6(\mathbf{Y})$  lies at the 83rd percentile of the new test statistic distribution, from which the extreme draws were .673 and .957—a clear improvement in fit.

## 6. CONCLUSIONS

We have presented a highly parameterized model that incorporates the structure of a complex dataset and supports the inferences of interest. Using LCVs with means related to observed covariates, we modeled CSS data consisting of a mixture of ordinal responses and an informative NA category, and related person and item characteristics to these responses. Although some inferences are directly comparable to those derived from more conventional analyses, other inferences—including assessments of the strength of evidence and those derived from latent processes—are facilitated by our hierarchical Bayesian approach.

We demonstrated how to assess the sensitivity of parameter estimates to individual cases and distributional assumptions. Posterior predictive model checks indicated that the initial model failed to address the nesting of questions within categories, but an augmented model fit well. This illustrates that model checking and modification is essential if a complex model is to improve on simpler approaches.

Although our model for the ordinal and NA responses is tailored to the characteristics of the DuPont CSSs, the model-building and inferential methods are much more general. The model could be altered easily to represent another theory about the decision process leading to an NA, by modifying the latent tree structure represented in Figure 1. Other observations, not necessarily involving missing data, can also be modeled using similar latent tree structures. Finally, our methods also can be generalized to hierarchical

models with a combination of dichotomous, ordinal, and continuous observations.

Should a company fit a simple model that ignores important features of the data but may be adequate for many inferences, or undertake the modeling effort required to build a complex model? The answer to this question depends on the financial impact of the decisions and the level of detail required. When a large number of expensive surveys have a standardized structure, the additional expense of building and applying a complex model is readily justifiable. Even when many inferences from the complex model agree with those from simpler analyses, rigorous testing of exploratory results can give management a higher degree of confidence. When the more sophisticated models yield new insights, the payoff in improved decision making can be tremendous.

#### APPENDIX: ALGORITHMS FOR BAYESIAN INFERENCE

We conduct inference by drawing samples from the joint posterior distribution  $[\theta|Y]$ , using Markov chain Monte Carlo simulation methods (Besag, Green, Higdon, and Mengersen 1995; Gelfand and Smith 1990; Gilks, Richardson, and Spiegelhalter 1995). Conditional distributions of coefficient vectors  $\beta_\alpha, \beta_\gamma$ , and  $\beta_\zeta$  and the corresponding variance components  $\sigma_\alpha^2, \sigma_\gamma^2$ , and  $\sigma_\zeta^2$  are given by standard normal and inverse-gamma theory.

The conditional distributions of the remaining parameters,  $\theta_1 = (\alpha, \gamma, \zeta), \beta_s, \beta_d$ , and  $c$ , involve the likelihood (2). Hence there is no closed-form sampling distribution for these parameters, although the conditional densities can be calculated up to a normalizing constant. Three alternative methods were used to sample these distributions: data augmentation (Albert and Chib 1993; Tanner and Wong 1987), a Metropolis step (Hastings 1970), and the Griddy-Gibbs sampler (Ritter and Tanner 1992). We present here the details of the first of the approaches. A detailed description of these approaches and efficiency comparisons have been given by Bradlow (1994).

Our approach uses a set of conditional distributions that are all in closed form, obtained by augmenting  $(\theta_1, \theta_2)$  with LCV "data"  $S, T$ , and  $D$ . The number of parameters is greatly increased, but sampling is very efficient. We elaborate briefly on this approach because it extends the method of Albert and Chib (1993) to handle the complex latent tree structure of the IZ model or similar models involving multiple latent variables.

The conditional distribution of  $s_{jk}$  combines its normal prior distribution with the likelihood and has three cases:

$$s_{jk} \sim \begin{cases} N^+(\mu_{s(jk)}, 1) & \text{if } y_{jk} \neq \text{NA} & \text{(Case 1)} \\ N(\mu_{s(jk)}, 1) & \text{if } y_{jk} = \text{NA}, C_L \leq t_{jk} < C_U, d_{jk} < 0 & \text{(Case 2)} \\ N^-(\mu_{s(jk)}, 1) & \text{otherwise} & \text{(Case 3),} \end{cases}$$

where  $N^+$  ( $N^-$ ) is the positive (negative) part of the normal distribution. In case 1, a non-NA is observed so the question must have been salient ( $s_{jk} \geq 0$ ). In case 2, an NA is observed,  $t_{jk}$  is in the IZ, and the person is not responsive ( $d_{jk} < 0$ ), so we would observe an NA, regardless of the value of  $s_{jk}$ . In case 3, we observed an NA and know that it did not appear due to indifference; hence the NA occurred because  $s_{jk} < 0$ ; that is, the question was not

salient. The conditional distributions for LCV matrices  $T$  and  $D$  are derived from truncated normals that can be deduced similarly from the latent tree of Figure 1.

Conditional on augmented data  $S, T$ , and  $D$ , the distributions of the remaining components of  $\theta$  (other than  $c$ ) are those of the hierarchical normal model. The conditional distribution of each cutoff  $c_r$  is uniform on the interval from the highest  $t_{jk}$  that must be below the cutoff to the lowest  $t_{jk}$  that must be above the cutoff, subject to the order constraints on cutoffs (relevant if a particular ordinal response never appears). For all cutoffs  $0 < r < R$  other than the bounds of the IZ or the fixed  $c_{r_0}$ ,

$$c_r \sim \text{unif}(\max(c_{r-1}, \max_{y_{jk}=r} t_{jk}), \min(c_{r+1}, \min_{y_{jk}=r+1} t_{jk})).$$

For the IZ bounds, the cutoffs must also be consistent with NA cases that are known to be due to indifference. Hence

$$c_{r_L-1} \sim \text{unif}(\max(c_{r_L-2}, \max_{\mathcal{I}} t_{jk}), \min(c_{r_L}, \min_{y_{jk}=r+1} t_{jk})),$$

where  $\mathcal{I} = \{(j, k) : y_{jk} = r_L - 1 \text{ or } (y_{jk} = \text{NA} \text{ and } s_{jk} > 0)\}$ , and the distribution of  $c_{r_U}$  is modified similarly.

Three parallel series were run to diagnose convergence (Gelman and Rubin 1992). Overdispersed starting values were chosen based on an initial run. Although the autocorrelation of the draws was high, each draw was fast making data augmentation a viable choice.

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